SYED MUHAMMAD SHARIQ

AI, Computer Vision & Backend Systems Engineer

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TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, Linux / Unix
Backend & Systems: FastAPI, WebSockets, REST APIs, Microservices, Asyncio, Pydantic v2, Google Vertex AI, Edge-TTS
Computer Vision & ML: PyTorch, YOLO (v8/v11), OpenCV, EasyOCR, Edge Device Optimization (TensorRT/ONNX), RTSP Video Streaming
DevOps & Tools: Docker, Docker Compose, Git, CI/CD (GitHub Actions), Google Cloud Run, Dataset Annotation (Roboflow, CVAT), SQLite

WORK EXPERIENCE

LA Consulting Corp — AI & Backend Systems Engineer Nov 2025 – Present | Karachi, Pakistan

- Architected backend inference microservices and automated multi-camera RTSP ingestion pipelines across 10,000+ custom annotated frames, cutting preprocessing turnaround by 35%.
- Benchmarked and deployed quantized YOLOv8 models on edge hardware, sustaining 30+ FPS across live camera feeds.
- Designed low-latency REST and WebSocket API endpoints for real-time tracking, integrated with automated CI validation.

Global Center for Rehumanization of Democracy (GCRD) — Speech & Data Researcher Sept 2026 – Present | Karachi, Pakistan

- Conduct acoustic analysis and data validation on human speech datasets to benchmark conversational nuance and transcription fidelity.
- Designed quality-assurance validation pipelines to filter phonetic inconsistencies and audio artifacts across speech-processing workflows.

State Bank of Pakistan (BSC) — Technology & Operations Internee June 2026 – August 2026 | Karachi, Pakistan

- Audited public procurement and enterprise compliance documentation in adherence with PPRA (Public Procurement Regulatory Authority) standards.
- Coordinated cross-departmental technology event operations, resource allocation, and internal systems documentation.

FEATURED PROJECTS

Intelligent Traffic Violation Detection System (Capstone Project) github.com/Shariq-duet/...

Tech: Python, PyTorch, YOLOv8, OpenCV, EasyOCR, Roboflow, Pandas

- Published 2 open-source benchmark datasets on Roboflow & Kaggle (helmet compliance and phone distraction) to address regional traffic data gaps.
- Built an automated multi-task enforcement pipeline detecting 5 infractions (helmet, seatbelt, phone, red-light jumping, and tailgating) achieving 91.4% mAP@50.
- Implemented heuristic crop selection (best_SS.py) scoring sharpness and contrast to filter motion blur before OCR plate extraction.

Rasta-AI — Voice-Guided Banking Assistant (Alibaba Cloud Hackathon National Finalist) github.com/Shariq-duet/rasta-ai

Tech: Python, WebSockets, Next.js, Ollama/LLMs, Urdu Speech-to-Text, Audio Processing

- Qualified in the top 300 teams out of 16,000+ submissions nationwide (top 1.8%); presented live at National Finals in Islamabad.
- Architected a bilingual (Urdu/English) voice navigation assistant designed to help low-literacy users interact with digital banking workflows.
- Built a real-time bi-directional WebSocket bridge synchronizing voice intents with Next.js UI elements, highlighting target buttons to guide users.

CallBridge — Autonomous Voice Negotiation Agent github.com/Shariq-duet/CallBridge

Tech: Python 3.12, FastAPI, WebSockets, AssemblyAI Voice Agent, Microsoft Edge-TTS, Pydantic v2, Docker

- Built for AssemblyAI Voice Agent Hackathon: developed a full-duplex conversational voice agent for outbound business negotiation calls.
- Engineered a hybrid speech stack pairing AssemblyAI turn detection with edge-tts neural audio and in-memory caching for low-latency responses.
- Implemented state machine safeguards with human-in-the-loop approval gates before confirming commercial terms or sensitive data.

Game Debugger — Automated Multi-Source Bug Triage Pipeline github.com/Shariq-duet/game-debugger-agent

Tech: Google Vertex AI, Gemini 2.5 Flash, Google Cloud Run, Pydantic v2, Discord API, Python

- Deployed a 3-phase automated agent on Google Cloud Run ingesting unstructured community bug reports across Discord, server logs, and CSVs.
- Built cross-source verification logic to reconcile conflicting reports and filter duplicates before generating reproducible Jira tickets and patch suggestions.

HONORS & COMPETITIONS

- **National Finalist (Top 1.8% of 16,000+ Teams):** Alibaba Cloud AI Hackathon Pakistan 2026. Fully sponsored presentation at Pakistan-China Friendship Center, Islamabad (Oct 2026).
- **Global Hackathon Submissions:** AssemblyAI Voice Agent Hackathon (CallBridge) & IBM Bob 2.0 Hackathon (Agentic LLMs with Anthropic Claude).

EDUCATION

Dawood University of Engineering & Technology (DUET) — Bachelor of Science in Computer Science (BSCS) 2023 – Expected May 2027 | Karachi, Pakistan